RELATIONAL DATABASE ANALYTICS PRESENTATION

IMDb Movies Data Analysis

ANALYST / PRESENTER
Abirami Thirunavukkarasu

PROGRAM
CDA Course

TEAM IDENTIFIER
PTID-CDA-SEP-26-11202

Dataset Focus
IMDb movie titles, directors & box-office data

SQL Mastery
JOINs, GROUP BY, aggregations & LIMIT queries

Business Insights
Bankable directors & popularity trends

Database Architecture & Table Structure

1. Directors Table

(Master Table)

Stores director profile details:

ID: Unique Primary Key

Name: Full name of director

Gender: Demographic code (1 = Female)

Department: Creative department

2. Movies Table

(Metrics Table)

Stores film performance metrics:

Title: Feature film title

Revenue: Total box office gross

Popularity: Audience engagement score

Vote average: User rating out of 10

3. Table Linking

(1-to-Many Link)

Relational key mapping:

Foreign Key: movies.director_id

Maps to directors.id Primary Key

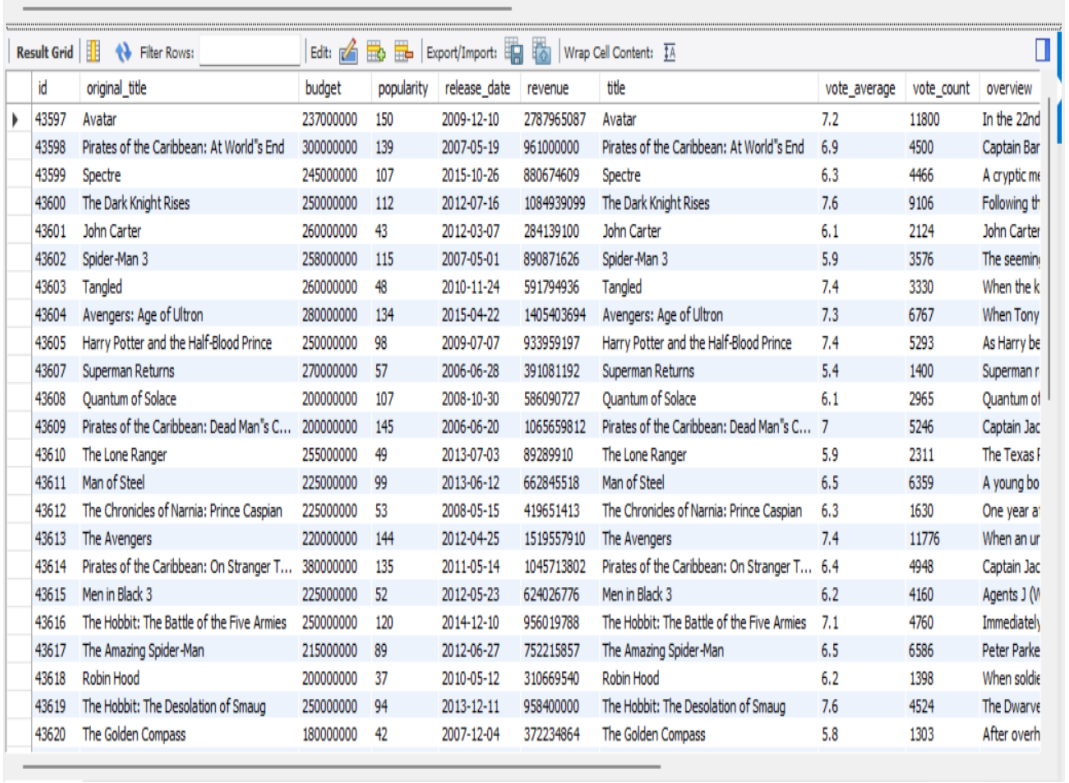
Cardinality: One director to multiple feature films.

Fetch all records from movies table

MySQL Query Code

```
# Can you get all data about movies?
SELECT * FROM movies;
```

MySQL Result Grid Output



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	original_title	budget	popularity	release_date	revenue	title	vote_average	vote_count	overview
43597	Avatar	237000000	150	2009-12-10	2787965087	Avatar	7.2	11800	In the 22nd
43598	Pirates of the Caribbean: At World's End	300000000	139	2007-05-19	961000000	Pirates of the Caribbean: At World's End	6.9	4500	Captain Bar
43599	Spectre	245000000	107	2015-10-26	880674609	Spectre	6.3	4466	A cryptic m
43600	The Dark Knight Rises	250000000	112	2012-07-16	1084939099	The Dark Knight Rises	7.6	9106	Following th
43601	John Carter	260000000	43	2012-03-07	284139100	John Carter	6.1	2124	John Carter
43602	Spider-Man 3	258000000	115	2007-05-01	890871626	Spider-Man 3	5.9	3576	The seemin
43603	Tangled	260000000	48	2010-11-24	591794936	Tangled	7.4	3330	When the k
43604	Avengers: Age of Ultron	280000000	134	2015-04-22	1405403694	Avengers: Age of Ultron	7.3	6767	When Tony
43605	Harry Potter and the Half-Blood Prince	250000000	98	2009-07-07	933959197	Harry Potter and the Half-Blood Prince	7.4	5293	As Harry be
43607	Superman Returns	270000000	57	2006-06-28	391081192	Superman Returns	5.4	1400	Superman r
43608	Quantum of Solace	200000000	107	2008-10-30	586090727	Quantum of Solace	6.1	2965	Quantum of
43609	Pirates of the Caribbean: Dead Man's C...	200000000	145	2006-06-20	1065659812	Pirates of the Caribbean: Dead Man's C...	7	5246	Captain Jac
43610	The Lone Ranger	255000000	49	2013-07-03	89289910	The Lone Ranger	5.9	2311	The Texas R
43611	Man of Steel	225000000	99	2013-06-12	662845518	Man of Steel	6.5	6359	A young bo
43612	The Chronicles of Narnia: Prince Caspian	225000000	53	2008-05-15	419651413	The Chronicles of Narnia: Prince Caspian	6.3	1630	One year a
43613	The Avengers	220000000	144	2012-04-25	1519557910	The Avengers	7.4	11776	When an ur
43614	Pirates of the Caribbean: On Stranger T...	380000000	135	2011-05-14	1045713802	Pirates of the Caribbean: On Stranger T...	6.4	4948	Captain Jac
43615	Men in Black 3	225000000	52	2012-05-23	624026776	Men in Black 3	6.2	4160	Agents J (M
43616	The Hobbit: The Battle of the Five Armies	250000000	120	2014-12-10	956019788	The Hobbit: The Battle of the Five Armies	7.1	4760	Immediately
43617	The Amazing Spider-Man	215000000	89	2012-06-27	752215857	The Amazing Spider-Man	6.5	6586	Peter Parke
43618	Robin Hood	200000000	37	2010-05-12	310669540	Robin Hood	6.2	1398	When soldie
43619	The Hobbit: The Desolation of Smaug	250000000	94	2013-12-11	958400000	The Hobbit: The Desolation of Smaug	7.6	4524	The Dwarve
43620	The Golden Compass	180000000	42	2007-12-04	372234864	The Golden Compass	5.8	1303	After overh

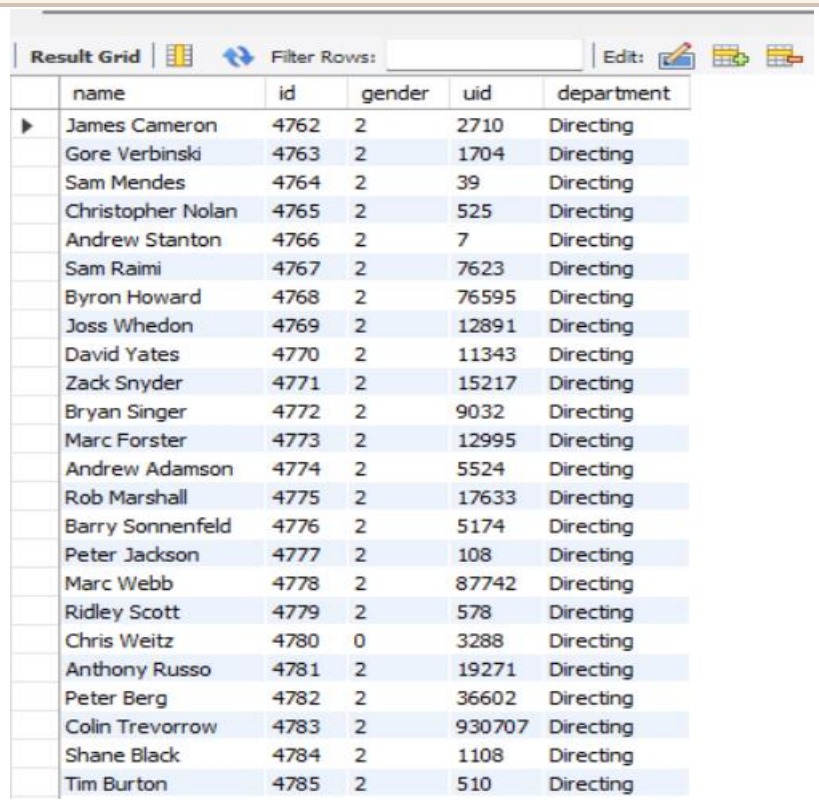
Description: This SQL statement extracts all columns and rows stored within the **movies** table without imposing any filter conditions.

Fetch all records from directors table

MySQL Query Code

```
# How do you get all data about directors?
SELECT * from directors;
```

MySQL Result Grid Output



Result Grid | Filter Rows: | Edit:

name	id	gender	uid	department
James Cameron	4762	2	2710	Directing
Gore Verbinski	4763	2	1704	Directing
Sam Mendes	4764	2	39	Directing
Christopher Nolan	4765	2	525	Directing
Andrew Stanton	4766	2	7	Directing
Sam Raimi	4767	2	7623	Directing
Byron Howard	4768	2	76595	Directing
Joss Whedon	4769	2	12891	Directing
David Yates	4770	2	11343	Directing
Zack Snyder	4771	2	15217	Directing
Bryan Singer	4772	2	9032	Directing
Marc Forster	4773	2	12995	Directing
Andrew Adamson	4774	2	5524	Directing
Rob Marshall	4775	2	17633	Directing
Barry Sonnenfeld	4776	2	5174	Directing
Peter Jackson	4777	2	108	Directing
Marc Webb	4778	2	87742	Directing
Ridley Scott	4779	2	578	Directing
Chris Weitz	4780	0	3288	Directing
Anthony Russo	4781	2	19271	Directing
Peter Berg	4782	2	36602	Directing
Colin Trevorrow	4783	2	930707	Directing
Shane Black	4784	2	1108	Directing
Tim Burton	4785	2	510	Directing

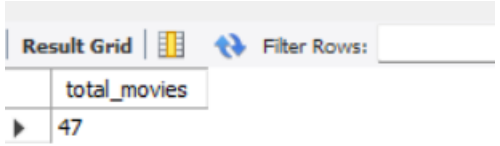
Description: Retrieves the complete master catalog of director records, including director primary IDs, full names, gender codes, and creative departments.

Total movies present in IMDb database

MySQL Query Code

```
# Check how many movies are present in IMDB.
SELECT COUNT(*) AS total_movies FROM movies;
```

MySQL Result Grid Output



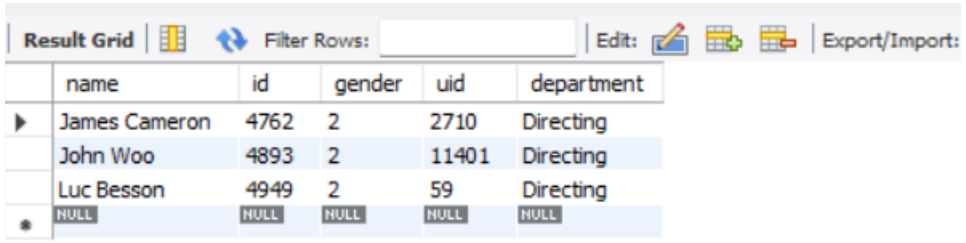
Description: Uses the aggregate function **COUNT(*)** with an explicit column alias **total movies** to compute the overall film volume in the dataset.

Search Specific Director Profiles Using IN Operator

MySQL Query Code

```
SELECT * FROM  directors
WHERE name IN ('James Cameron', 'Luc Besson', 'John Woo');
```

MySQL Result Grid Output



name	id	gender	uid	department
James Cameron	4762	2	2710	Directing
John Woo	4893	2	11401	Directing
Luc Besson	4949	2	59	Directing
NULL	NULL	NULL	NULL	NULL

Description: Applies the set membership **IN ('James Cameron', 'Luc Besson', 'John Woo')** operator to efficiently fetch exact profile details for these specified directors.

Find directors whose names start with 'S'

MySQL Query Code

```
SELECT * FROM directors
WHERE name LIKE 'S%';
```

MySQL Result Grid Output

Result Grid | Filter Rows: | Edit: | Export/Import:

name	id	gender	uid	department
Sam Mendes	4764	2	39	Directing
Sam Raimi	4767	2	7623	Directing
Shane Black	4784	2	1108	Directing
Steven Spielberg	4799	2	488	Directing
Stephen Sommers	4815	2	7775	Directing
Shawn Levy	4842	2	17825	Directing
Steve Hickner	4852	2	44113	Directing
Simon Wells	4855	2	21879	Directing
Steven Soderbergh	4909	2	1884	Directing
Simon West	4930	2	12786	Directing
Stefen Fangmeier	4931	0	25453	Directing
Spike Jonze	4932	2	5953	Directing
Steve Martino	4943	2	71729	Directing
Sergei Bodrov	4952	0	130938	Directing
Sydney Pollack	4965	2	2226	Directing
Sylvester Stallone	4992	2	16483	Directing
Seth Gordon	4997	2	71600	Directing
Scott Derrickson	5004	2	55499	Directing
Stephen Hopkins	5008	2	2042	Directing
Steven Brill	5013	2	32593	Directing

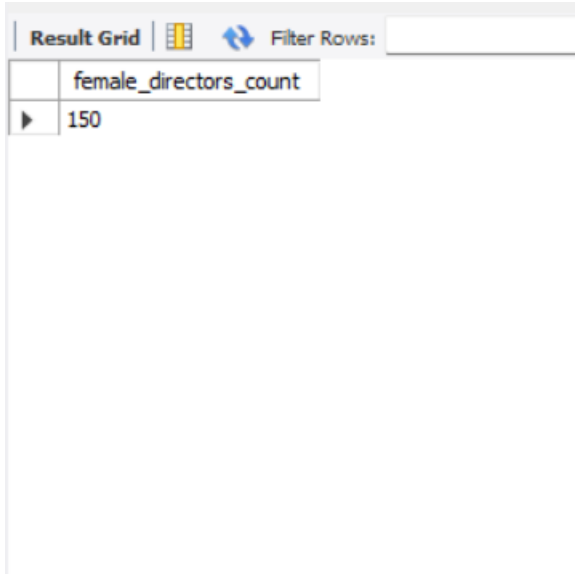
Description: Uses SQL wildcard pattern matching with **LIKE 'S%'** to isolate director records where the director's name begins with the letter 'S'.

Count total female directors in database

MySQL Query Code

```
SELECT COUNT(*) AS female_directors_count FROM directors
WHERE gender = 1 ;
```

MySQL Result Grid Output



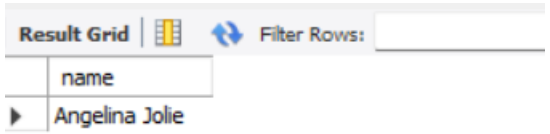
Description: Aggregates director rows filtered by demographic code **WHERE gender = 1** to derive total female director representation.

Name of the 10th female director

MySQL Query Code

```
SELECT name FROM directors
WHERE gender = 1
ORDER BY id
LIMIT 1 OFFSET 9;
```

MySQL Result Grid Output



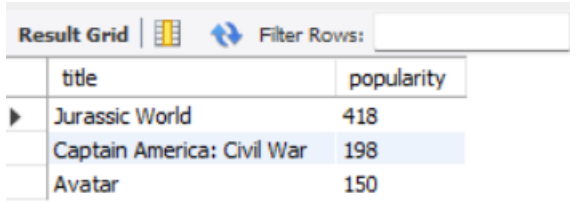
Description: Combines pagination clause **LIMIT 1 OFFSET 9** with **ORDER BY id** to skip the first 9 records and pinpoint the 10th female director.

3 most popular movies

MySQL Query Code

```
SELECT title, popularity FROM movies
ORDER BY popularity DESC
LIMIT 3;
```

MySQL Result Grid Output



title	popularity
Jurassic World	418
Captain America: Civil War	198
Avatar	150

Description: Sorts the dataset in descending order using **ORDER BY popularity DESC** and caps output to top 3 using **LIMIT 3**.

3 most bankable movies (Highest Revenue)

MySQL Query Code

```
SELECT title, revenue FROM movies
ORDER BY revenue DESC
LIMIT 3;
```

MySQL Result Grid Output

Result Grid | Filter Rows:

title	revenue
Avatar	2787965087
Titanic	1845034188
The Avengers	1519557910

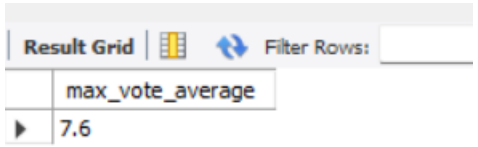
Description: Ranks box-office revenue generation via **ORDER BY revenue DESC LIMIT 3** to identify the highest grossing commercial films.

Maximum vote average since Jan 1, 2000

MySQL Query Code

```
SELECT MAX(vote_average) AS max_vote_average FROM movies
WHERE release_date >= '2000-01-01';
```

MySQL Result Grid Output



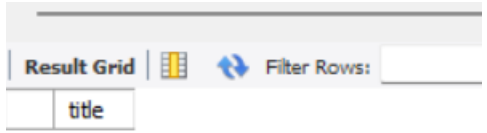
Description: Filters release date using **WHERE release_date >= '2000-01-01'** and extracts maximum rating using **MAX(vote_average)**.

Retrieve Movies Directed by Brenda Chapman

MySQL Query Code

```sql
# Which movies were directed by Brenda Chapman
SELECT m.title FROM movies m
JOIN directors d ON m.director_id = d.id
WHERE d.name = 'Brenda Chapman';
```

MySQL Result Grid Output



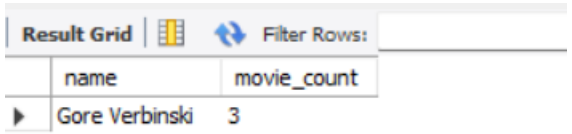
Description: The empty output indicates that no movie records directed by Brenda chapman are stored in the database.

Director who directed the most movies

MySQL Query Code

```sql
SELECT d.name, COUNT(m.id) AS movie_count
FROM directors d
JOIN movies m ON d.id = m.director_id
GROUP BY d.id, d.name
ORDER BY movie_count DESC
LIMIT 1;
```

MySQL Result Grid Output



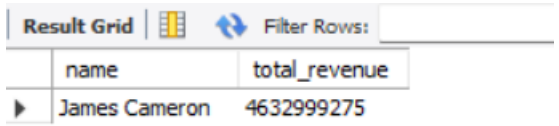
Description: Groups movie counts per director using **GROUP BY d.id** and applies **ORDER BY total movies DESC LIMIT 1** to isolate the highest-output director.

Which director is the most bankable?

MySQL Query Code

```sql
SELECT d.name, SUM(m.revenue) AS total_revenue
FROM directors d
JOIN movies m ON d.id = m.director_id
GROUP BY d.id, d.name
ORDER BY total_revenue DESC
LIMIT 1;
```

MySQL Result Grid Output



name	total_revenue
James Cameron	4632999275

Description: Aggregates cumulative revenue per director using **SUM(m.revenue)** and identifies the single most financially bankable director.

Project Summary & Key Analytical Takeaways

1. Relational Integrity & Key Joins

Accurately mapped **movies.director_id** to **directors.id** across multi-table INNER JOINs to combine film metrics with director profiles.

2. Aggregations & Financial Metrics

Evaluated total film output (**COUNT**), maximum ratings (**MAX**), and cumulative director gross revenue (**SUM**).

3. Ranking & Filtering Precision

Implemented precise set filtering (**IN**, **LIKE**), ordering (**ORDER BY DESC**), and result capping (**LIMIT / OFFSET**).